

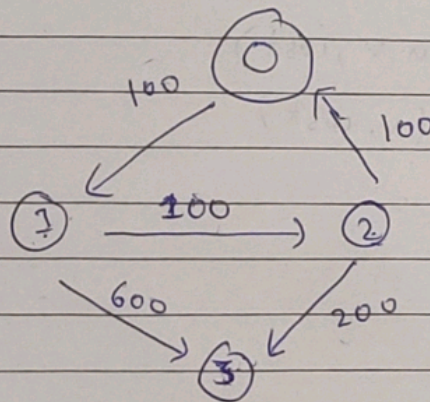
EXPERIMENT - 7

You are given n cities connected by some number of flights. where flight $[i] = [\text{from}, \text{to}, \text{price}]$ indicates that there is a flight from city from to city to with cost price.

You are also given three integers, src , dst and k , return the cheapest price from src to dst with at most k stops. If there is no such that, return -1 .

Input : $n=4$, flight = $[[0,1,100], [1,2,100], [2,0,100], [0,3,600], [2,3,200]]$, $\text{src}=0$, $\text{dst}=3$, $k=1$

Output = 700



Approach : we will use modified BFS here (Kahn style) and keep track of :

- node
- cost
- stops

Use Queue and distance array.

Teacher's Signature :

Explore level by level (stops) and relax edges only if cost improves.

1. Build adj list (node $\rightarrow$ neighbour, cost)
2. Use queue : { node, cost, stops }
3. Maintain dist [] $\rightarrow$ minimum cost to reach node.
4. Traverse until stops $\leq k$
5. Update only if better cost

function cheapestFlight (n, flights, src, dst, k)

adj = adjacency list

for each flight (u, v, cost)

adj[u].push (v, cost)

dist [] = ∞

dist [src] = 0

queue = { src, 0, 0 }

while (! q.empty())

{

pop = node, cost, stops

if (stops > k) continue // skip

for each (neighbour, weight)

2 $\rightarrow$

3 $\rightarrow$

relax

if cost + weight < dist [neighbour] // Relax

dist [neighbour] = cost + weight ;

q.push (neighbour, new cost, step + 1);

if (distance[dst] == ∞);
return -1

return dist[dst];

Time complexity $O(E)$

BFS traversal with relaxation

Space Complexity $O(V + E)$